A-A-51720A
19 June 1989
SUPERSEDING
A-A-0051720 (DM)
17 November 1986

COMMERCIAL ITEM DESCRIPTION

CARRIER, AMALGAM

The General Services Administration has authorized the use of this Commercial Item Description.

Salient Characteristics:

This commercial item description covers a lever type amalgam carrier suitable for delivering a freshly mixed dental amalgam into a prepared tooth cavity. The carrier shall be of the following styles and sizes.

Style A - Single-ended

Size 1 - Regular 2.0 mm

Size 2 - Large 2.5 mm

Style B - Double-ended

Size 1 - One end regular, 2.0 mm, and opposing end large, 2.5 mm

Size 2 - One end regular, 2.0 mm, and opposing end mini, 1.4 mm

The carrier shall receive, carry and deliver a dental amalgam into a prepared tooth cavity during dental restoration procedures. The components shall be properly oriented to each other to provide for a smooth operating instrument without binding and side play. In operation, the plunger shall protrude beyond the barrel when lever is fully depressed. The spring shall return the moving components to the original rest position. The distal opening in the barrel shall extend for at least 5/32 inch. The plunger shall fit the barrel with minimum clearance.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:00Z
Check the source to verify that this is the current version before use.

The style A, size 1 carrier shall be single-ended with a regular size (2.0-2.2 mm) barrel. The style A, size 2 carrier shall be single-ended with a large size (2.5-2.8 mm) barrel. The style B, size 1 carrier shall be double-ended with one regular size (2.0-2.2 mm) barrel and one large size (2.5-2.8 mm) barrel. The style B, size 2 carrier shall be double-ended with one regular size (2.0-2.2 mm) barrel and one mini size (1.4-1.5 mm) barrel.

Material. The barrel, lever, and lever spring shall be fabricated from martensitic corrosion resisting steel which is in accordance with ASTM Designation F 899.

Finish. The plunger surface shall be smooth and the interior surface of the barrel shall have a highly polished finish, to prevent jamming of the hardened alloy in the barrel.

Hardness. The lever shall have a Rockwell hardness of C30 to C35, and the lever spring components shall have a Rockwell hardness of C47 to C52 after proper heat treatment and stress relief.

Corrosion resistance. Finished corrosion resisting steel components shall be passivated. There shall be no copper plating on the carrier and no signs of corrosion on all smooth exposed surfaces of the carrier when subjected to the copper sulfate and boil tests as specified herein.

Operation. The carriers used in the boil test shall not fail to operate during 25 simulated cycles of operation. Smoothness and non-binding operation shall be the same as new untested carriers. Examination of operating parts shall show no evidence of cracking, breakage or deformation due to the simulated action.

Amalgam expulsion. The carriers shall completely expel the pellet with no amalgam remaining in the barrel when tested as specified herein.

Identification marking. Instruments of foreign manufacture shall be permanently and legibly marked with the country of origin. The carrier shall be permanently identified with the name or registered trademark of the supplier.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:00Z
Check the source to verify that this is the current version before use.

Workmanship. The carrier shall be free from cracks, holes, pitting, burrs and excessive sharpness which will detract from its appearance or impair its serviceability.

Unit. Each (EA). One carrier, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Inspection shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The unit of product for examination purposes shall be one carrier. Unless otherwise specified, sampling for examination shall be at inspection level II, AQL 2.5 (percent defective). Dimensions shall be examined at inspection level S-1, AQL 1.0 (percent defective).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:00Z
Check the source to verify that this is the current version before use.

Defects

Carriers not free from cracks, burrs, or excessively sharp edges.
Dimensions not as specified.
Finish not as specified.
Plunger fails to protrude from barrel when lever is depressed.
Lever spring fails to return lever to barrel to the same rest position.
Instrument fails to operate smoothly when depressing and releasing lever (without sticking, binding, wobbling).
Component parts fit together with excessive looseness.
Assembled components not properly aligned.
Identification markings incorrect, incomplete, illegible or not permanent.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more test characteristics. The unit of product for test purposes shall be one carrier. Sampling for tests shall be conducted at inspection level S-1, AQL 1.0 (percent defective).

Tests.

Hardness. Testing for hardness shall be conducted in accordance with ASTM Designation E-18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

Corrosion resistance. The resistance to corrosion of the carrier's components shall be determined by the copper sulfate and boil tests of ASTM Designation F 1089.

Pellet expulsion. Carriers from the boil test shall be filled with amalgam made from a Tytin 600 mg or 800 mg capsule according to the manufacturer's instructions. The amalgam in the barrel shall be allowed to harden undisturbed for three hours. The carriers shall then be operated to expel the pellet by hand pressure on the handle with the lever resting on its lip and the barrel centered inside the neck of an inverted alloy funnel or similar device.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:00Z
Check the source to verify that this is the current version before use.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pounds units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:00Z
Check the source to verify that this is the current version before use.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification
6520-00-405-0765	CARRIER, AMALGAM, Lever Type, Double-ended, Large and Regular Barrels
6520-00-511-2050	CARRIER, AMALGAM, Lever Type, Single-ended, Regular Barrel
6520-00-955-5275	CARRIER, AMALGAM, Lever Type, Single-ended, Large Barrel
6520-01-261-5971	CARRIER, AMALGAM, Lever Type, Double-ended, Mini and Regular Barrels

Ordering data. Intermediate/exterior package quantities, labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2270
LOCATION: NCR DISK 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:00Z
Check the source to verify that this is the current version before use.